Map of MALDER results

2025-03-19

```
res_dir <- "../results/s07_b_MALDER_map/"
```

Load data

```
dates_full <- readRDS("../results/s07_MALDER/malder_results_all.RDS")
```

This script will create a map showing admixture timing with Mozambique as an African source population and Borneo as an Asian source population.

Subset to reference popualtions of interest

```
dates <- dates_full %>%
  filter(
    `ref A` == "Borneo" &
      `ref B` == "Mozambique_Chopi"
  )
```

Load locations

```
kml_data_full <-
  readRDS("../data/raw/data_input/sampling_locations_with_pierron.rds")
```

Merina A and Merina B are the same population, sampled from the same area. To show two points for them on the map, in order to represent the different admixture dates estimated for each group, we create a point for Merina B by adjusting the x coordinate to be slightly south of Merina A.

```
group <- c("MerinaA", "MerinaB")
x <- c(519090.2, 519090.2)
y <- c(834836.2, 810000.0)
sample <- c(0, 0)

merina_coords <- data.frame(Name = group, x = x, y = y, sample = sample)

merina_coords <- merina_coords %>%
  st_as_sf(coords = c("x", "y"), crs = 29702) %>%
```

```
  mutate(
    x = st_coordinates(.)[, 1],
    y = st_coordinates(.)[, 2]
  )

kml_data <- kml_data_full %>%
  mutate(
    x = st_coordinates(.)[, 1],
    y = st_coordinates(.)[, 2]
  )

# rbind the two datasets
kml_data_all <- rbind(kml_data, merina_coords)
# remove Merina
kml_data <- kml_data_all %>%
  filter(Name != "Merina")
```

Combine admixture dates and sampling locations

```
# match kml_data$Name and dates rownames
matching <- match(kml_data$Name, as.character(dates$test_pop))

# add admixture date and std_err to kml_data
kml_data$date <- dates[matching, "admixture_date"]
kml_data$std_err <- dates[matching, "CI"]

kml_data$sample <-
  ifelse(kml_data$Name %in% c(
    "Merina", "MerinaA", "MerinaB",
    "Sakalava_Tsimihety", "Betsimisaraka",
    "Betsileo", "Diego"
  ), 0, 1)

kml_data <- kml_data[order(kml_data$Name), ]

kml_data$shape <- c(1, 3, 7, 15, 8, 17, 17, 17, 17, 17, 17, 17, 19, 21, 23)

# add admixture date in years (gen time 30 years)
kml_data$years <- paste(
  round(kml_data$date * 30, 0),
  "\u00b1",
  round(kml_data$std_err * 30, 0), "BP"
)


labels_data <- kml_data %>%
  group_by(Name) %>%
  reframe(
    x = mean(x),
    y = mean(y),
    date = years,
```

```
    geometry = st_centroid(st_union(geometry))
  )

# Mikea has multiple locations, remove duplicates and leave one row to label
labels_data <- labels_data[!duplicated(labels_data$Name), ]
```

Create base map

```
target <- c(
  "Madagascar", "Mozambique", "Comoros", "Mauritius", "Seychelles",
  "Mayotte", "United Republic of Tanzania", "Malawi", "Zimbabwe",
  "South Africa", "Eswatini", "Zambia", "Lesotho", "Somalia",
  "Kenya", "Ethiopia", "Uganda", "Rwanda", "Burundi", "Djibouti",
  "Eritrea", "South Sudan", "Botswana", "Namibia", "Angola",
  "Democratic Republic of the Congo", "Central African Republic",
  "Republic of the Congo", "Equatorial Guinea", "Cameroon"
)

# Create an inset of Madagascar
madagascar_inset <- ne_countries(
  country = target,
  type = "map_units",
  scale = 50
)

madagascar_dissolved <- madagascar_inset %>%
  st_union() %>%
  st_as_sf()
```

Plot dates on base map

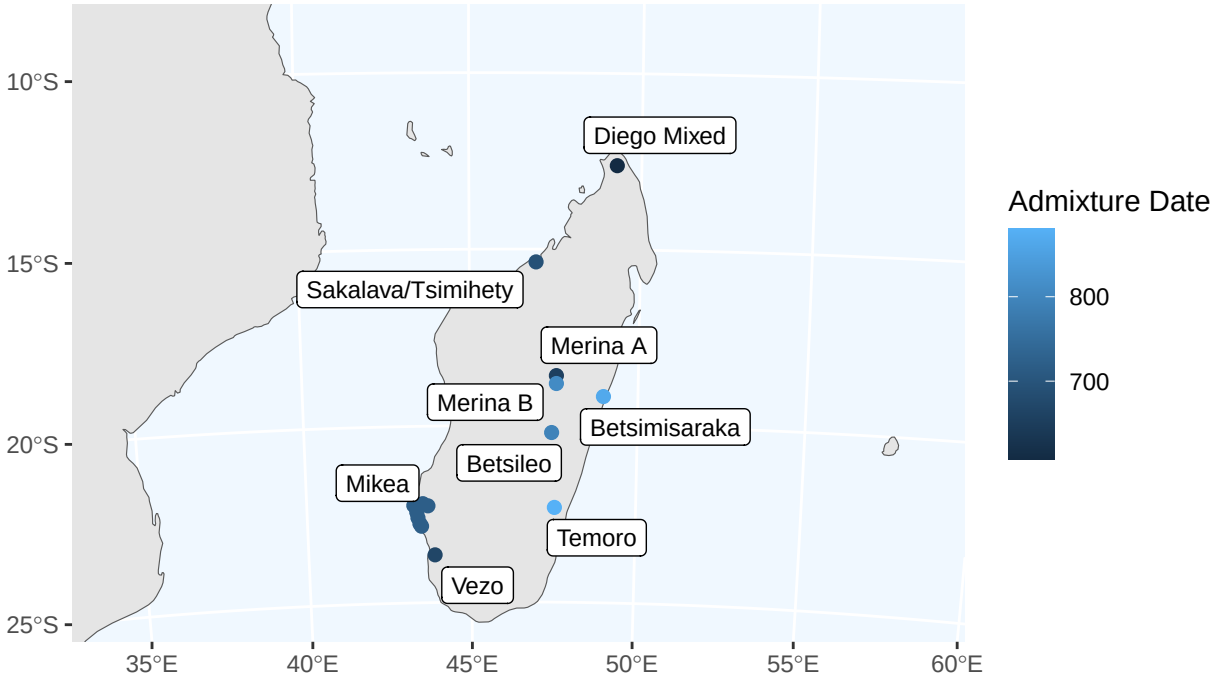
```
kml_data$yrs <- as.numeric(unlist(tstrsplit(kml_data$years, " ", keep = 1)))

# Rename
kml_data$Name[kml_data$Name == "Sakalava_Tsimihety"] <- "Sakalava/Tsimihety"
kml_data$Name[kml_data$Name == "Diego"] <- "Diego Mixed"
kml_data$Name[kml_data$Name == "MerinaA"] <- "Merina A"
kml_data$Name[kml_data$Name == "MerinaB"] <- "Merina B"

labels_data <- kml_data %>%
  group_by(Name) %>%
  summarise(
    x = mean(x),
    y = mean(y),
    geometry = st_centroid(st_union(geometry))
  )

ggplot() +
```

```
  geom_sf(data = madagascar_dissolved) +
  geom_sf(
    data = sf::st_as_sf(kml_data),
    aes(color = yrs, fill = yrs),
    size = 2
  ) +
  coord_sf(
    crs = st_crs(29702),
    xlim = c(-1000000, 1800000),
    ylim = c(1, 2000000),
    expand = FALSE
  ) + # Laborde projection
  theme(
    panel.background = element_rect(fill = "#F0F8FF", color = NA)
  ) +
  labs(
    colour = "Admixture Date",
    fill = "Admixture Date",
    x = NULL, y = NULL
  ) +
  geom_label_repel(
    data = labels_data,
    aes(x = x, y = y, label = Name),
    size = 3, segment.color = "transparent"
  ) +
  guides()
```



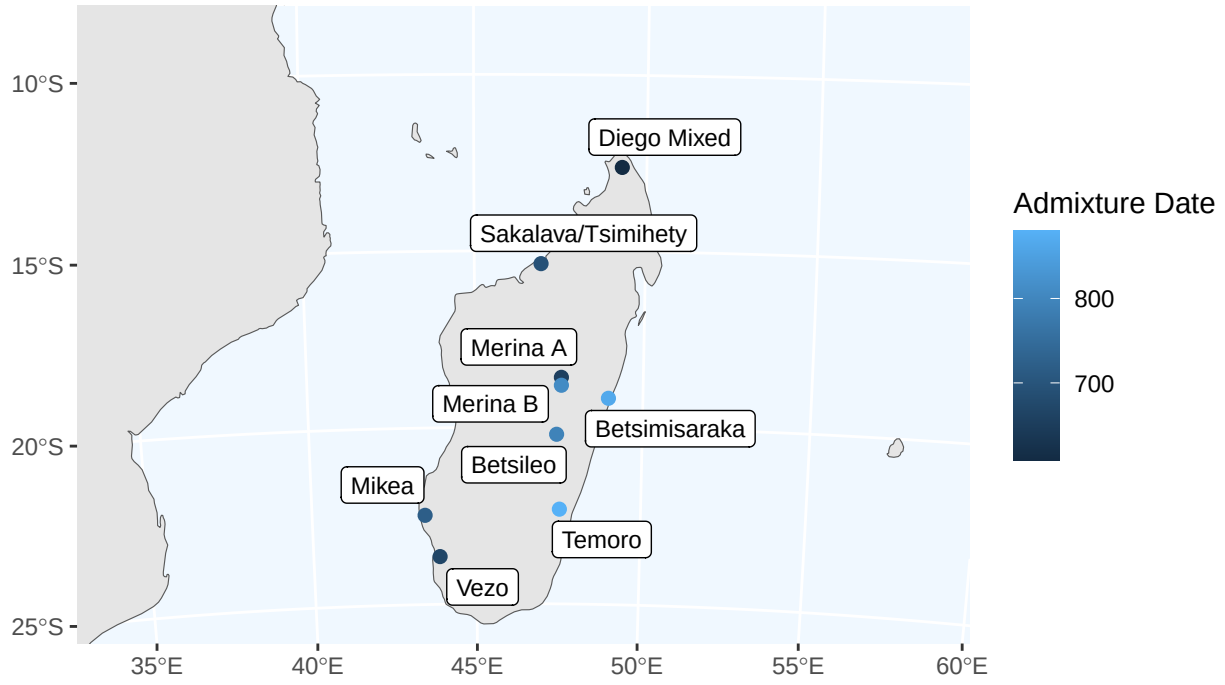

```
ggsave(paste0(res_dir, "malder_dates_gradient.jpg"),
  width = 7, height = 4, dpi = 300
)
```

Plot dates on base map - single point for Mikea

```
kml_data_single_site <- kml_data %>%
  group_by(Name) %>%
  summarize(geometry = st_union(geometry)) %>%
  st_centroid()
kml_data_single_site$yrs <-
  kml_data$yrs[match(kml_data_single_site$Name, kml_data$Name)]

plt <- ggplot() +
  geom_sf(data = madagascar_dissolved) +
  geom_sf(
    data = sf::st_as_sf(kml_data_single_site),
    aes(color = yrs, fill = yrs),
    size = 2
  ) +
  coord_sf(
    crs = st_crs(29702),
    xlim = c(-1000000, 1800000),
```

```
    ylim = c(1, 2000000),
    expand = FALSE
  ) + # Laborde projection
  theme(
    panel.background = element_rect(fill = "#F0F8FF", color = NA)
  ) +
  labs(
    colour = "Admixture Date",
    fill = "Admixture Date",
    x = NULL, y = NULL
  ) +
  geom_label_repel(
    data = labels_data,
    aes(x = x, y = y, label = Name),
    size = 3, segment.color = "transparent"
  ) +
  guides()
plt
```




```
ggsave(paste0(res_dir, "malder_dates_gradient_single_site.jpg"),
  width = 7, height = 4, dpi = 300
)
```

```r
pdf(paste0(res_dir, "figure_4_MALDER.pdf"), width = 7, height = 4, useDingbats = FALSE)
plt
dev.off()
```

```
## pdf
##   2
```
